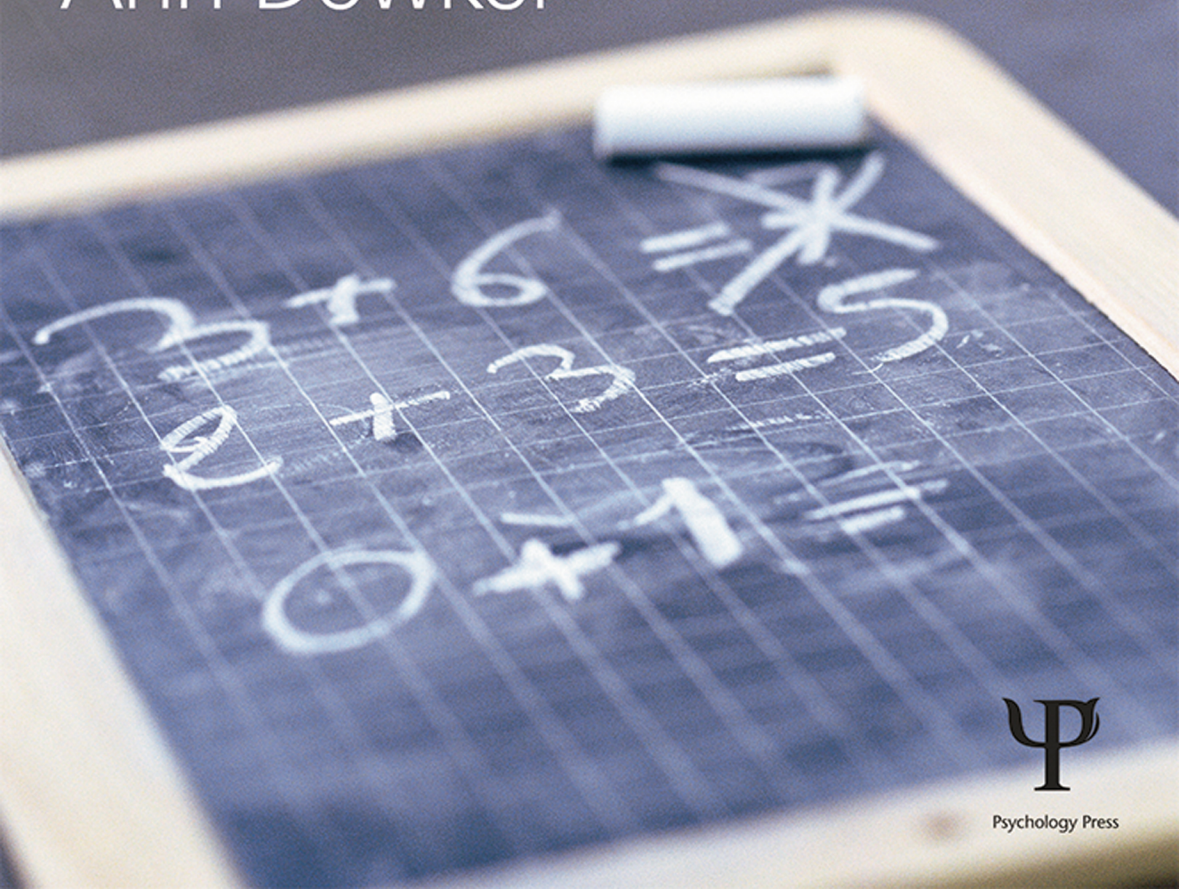
Second Edition

Individual Differences in Arithmetic

Implications for Psychology, Neuroscience and Education

Ann Dowker



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INDIVIDUAL DIFFERENCES IN ARITHMETIC

Arithmetic is still hugely important in many aspects of modern life, but our personal attitudes to it differ greatly. Many people struggle with the basic principles of arithmetic, whilst others love it and feel confident in their arithmetical abilities. Why are there so many individual differences in people's performance in, and feelings about, arithmetic?

Individual Differences in Arithmetic explores the idea that there is no such thing as arithmetical *ability*, only arithmetical *abilities*. The book discusses several important components of arithmetic, from counting principles and procedures to arithmetical estimation, alongside emotional and cognitive components of arithmetical performance. This edition has been extensively revised to cover the latest research, including recent cross-cultural and cross-linguistic research, the development of new interventions for children with difficulties and studies of early foundations of mathematical abilities.

Drawing on developmental, educational, cognitive and neuropsychological studies, this book will be essential reading for all researchers of mathematical cognition. It will also be of interest to educators and other professionals working with individuals with arithmetic deficits.

Ann Dowker is a University Research Lecturer in the Department of Experimental Psychology, University of Oxford. She is the lead researcher on the Catch Up Numeracy intervention project.

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Ann Dowker

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To the memories of Bob Hiorns and Neil O'Connor

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'ARITHMETIC'

Carl Sandburg

Arithmetic is where numbers fly like pigeons in and out of your head.
Arithmetic tells you how many you lose or win if you know how many you had before you lost or won.
Arithmetic is seven eleven all good children go to heaven–or five six bundle of sticks.
Arithmetic is numbers you squeeze from your head to your hand to your pencil to your paper till you get the answer.
Arithmetic is where the answer is right and everything is nice and you can look out of the window and see the blue sky – or the answer is wrong and you have to start all over and try again and see how it comes out this time.
If you take a number and double it and double it again and then double it a few more times, the number gets bigger and bigger and goes higher and higher and only arithmetic can tell you what the number is when you decide to quit doubling.
Arithmetic is where you have to multiply – and you carry the multiplication table in your head and hope you won't lose it.
If you have two animal crackers, one good and one bad, and you eat one and a striped zebra with streaks all over him eats the other, how many animal crackers will you have if somebody offers you five six seven and you say No no no and you say Nay nay nay and you say Nix nix nix?
If you ask your mother for one fried egg for breakfast and she gives you two fried eggs and you eat both of them, who is better in arithmetic, you or your mother?

"Arithmetic" from The Complete Poems of Carl Sandburg, *Revised and Expanded Edition. Copyright © 1970, 1969 by Lillian Steichen Sandburg, Trustee. Reprinted by permission of Houghton Mifflin Harcourt Publishing Company. All rights reserved.*

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INTRODUCTION

What this book is

This book is a review of the extent and nature of individual differences in arithmetic in childhood and after. This book will discuss several important components of arithmetical cognition and performance, and will discuss individual differences in these components: focusing in particular on children, but also drawing on studies of arithmetical cognition in healthy adults and in neuropsychological patients.

There is a great deal of concern in many countries about standards in numeracy (e.g. BIS, 2011; Ginsburg & Pappas, 2007). An example of this concern is the establishment of the National Numeracy organization in the UK in 2012, dedicated to improving competence in and attitudes to arithmetic. However, it may be difficult to ameliorate or prevent numeracy difficulties without a greater understanding of their nature.

The main argument of this book is that there is no such thing as arithmetical ability: only arithmetical abilities. Converging evidence from studies of normally developing children, of adults from the general population, of neuropsychological patients, of others with arithmetical disabilities and of people with exceptional arithmetical talent, and from recent brain imaging studies indicates that arithmetical cognition is made up of many components. Marked individual differences in these components, and discrepancies between these components within an individual child or adult, can be readily observed.

The book will discuss several important components of arithmetic and evidence for individual differences in these components. The components to be discussed include counting principles and procedures; knowledge of arithmetical facts procedures; the understanding of arithmetical principles and ability to use them to devise new, non-standard strategies; arithmetical estimation; and the

ability to deal with sums presented in terms of concrete objects, numbers and word problems, and to translate between the different versions. The book will discuss the contributions that internal (the brain) and external (environmental, cultural and linguistic) factors make to such individual differences. It will also be emphasized that arithmetical performance has emotional as well as cognitive components, and the book will discuss studies of emotions and attitudes concerning mathematics. The book will close with a discussion of the educational implications of these findings; of some remedial arithmetic programmes for those with arithmetical difficulties, and finally with overall conclusions from the material discussed in the book.

A major aim: bringing together findings from different times, places and subjects

Arithmetical development and arithmetical thinking have been important topics of study for many years, in many parts of the world and from the point of view of many disciplines. These topics are of concern to teachers, educational policymakers, psychologists, neuroscientists and mathematicians, and to all who need to learn arithmetic and to deal with numbers.

Much less work has been carried out on arithmetical development than, for example, the development of reading. At the same time, much more work has been done on the subject than is often realized. This is in part because of lack of transmission of information across time, across place and across discipline.

With regard to transmission across time, there is a significant body of early work, which is sometimes ignored nowadays. In the past, theories and findings were not always transmitted as efficiently as might be the case today, and it is easy for people of the 21st century, when examining something as seemingly topical as mathematics education, to assume that they are coming new to the topic. It can be startling to discover, for example, that the need for individualized intervention programmes for children with arithmetical difficulties was already being pointed out in the 1920s, that 19th-century educational policymakers were already debating the relative importance of rote learning and conceptual understanding (Brown, 2001; McIntosh, 1977), and that Maria and Richard Edgeworth were emphasizing the importance of language to children's understanding of arithmetic in 1798.

With regard to transmission across place, researchers are not always aware of what is going on in other countries. This is particularly true when language barriers are involved; but even communication between Britain and North America is sometimes imperfect. The bibliographies in British and American books on mathematical development often show remarkably little overlap.

With regard to transmission of information across disciplines, there is still a tendency for research in different disciplines to proceed independently, so that neuroscientists, developmental psychologists and educationists may not even be aware of mutually relevant work: a situation which limits the scope of such research. Communication between researchers, teachers and other practitioners, and policymakers is still more inadequate.

Thus, a key aim of the present book is to bring together work that is relevant to individual differences in arithmetic from different times, different places and across the disciplines of psychology, education and neuroscience.

What this book is not

There are several potentially interesting issues that are beyond the scope of this book:

It is not about the whole of mathematics. It deals specifically with arithmetic, which is only one of the many aspects of mathematics: for example, it does not deal with geometry.

It does not attempt to provide a comprehensive definition of arithmetic and its components from a mathematical, philosophical or educational point of view.

It does not investigate individual differences in arithmetic mainly from the point of view of the 'nature–nurture debate', though there will be some references to studies involving genetics.

It is not concerned with the issue of how arithmetic might have evolved.

It is not an instruction manual for parents or teachers in specific techniques of improving children's number skills, though it is hoped that the material in the book, and especially in the chapter on interventions, may provide some guidance in this area. A list of practical guides to helping individuals with mathematical difficulties will be given at the end of the book.

A note regarding terminology

Such terms as 'skills' and 'abilities' will at times be used to refer to proficiency in arithmetic, its components and other domains. The use of such terms does not imply assumptions about the genetic or environmental causation of such 'abilities', or whether they are innate or learned. Unfortunately, there are no common terms in this area, which have not become associated to some degree with nature/nurture studies. If there were such neutral terms, these would be used.

The main themes of the book

The main themes of the book are as follows:

1 Arithmetical ability is not unitary, but is made up of many different types of process. These can be grouped into several categories (e.g. procedural, factual and conceptual), but each of the categories has numerous subcomponents. For example, procedural knowledge may include written, oral and concrete calculation procedures, and includes the procedures applicable to different arithmetical operations. Factual knowledge will include facts corresponding to different operations, e.g. addition facts and multiplication facts, and also the names given to different numbers and operations. Conceptual

knowledge is an even broader category, including the understanding of the meaning of word problems; approximate arithmetic; and arithmetical principles such as commutativity, associativity and distributivity.

2 Some subcomponents will be easier for children than others; but there can be strong discrepancies, in either direction, between almost any two subcomponents. For example, though counting is generally easier than word problem solving, there are some children who have more difficulty with counting procedures than with word problem solving (see Chapter 4).

3 There are many parallels between the discrepancies that can be observed in 'normal' children and adults and those that can be observed in individuals with arithmetical deficits resulting from brain damage. This should not be seen as implying either that the deficits in brain-damaged people are not real, or that discrepancies in the general population must be due to subtle brain damage or abnormal brain functioning. Rather, performance in each of the different subcomponents of arithmetic can best be seen as lying on a continuum, with extreme talent at one end and extreme deficit (sometimes, but not always, due to brain damage) at the other.

4 Levels of performance in different subcomponents are influenced by a multitude of interacting factors: brain-based (both innate and acquired), social, cultural and educational. Any statement that arithmetical ability is purely the product of a single factor is oversimplified.

5 Emotional factors make a very important contribution to arithmetical performance. Some people like arithmetic very much, while a worryingly large number of people experience 'math anxiety' or even 'math phobia'. There are many contributory factors to such negative attitudes to arithmetic; but I would suggest that one of these is the tendency of children themselves and of their parents and teachers to regard arithmetic as unitary and to assume that difficulty with any one aspect of arithmetic means that one is 'no good at math'.

6 Arithmetical difficulties can be ameliorated considerably by intervention, especially interventions that are based on the fact that arithmetic is made up of numerous subcomponents and that take account of the specific strengths and weaknesses of individuals.

1

INDIVIDUAL DIFFERENCES IN ARITHMETIC

Children, adults, males, females, weaknesses and talents

> Most people can do maths when they're 4, if they start school when they're 4. And when they're grown up, some people can do maths and some can't.
>
> *(Tom, aged 6)*

It is well known that individual differences in arithmetical performance are very marked in both children and adults. Kaufmann and Nuerk (2005) claimed that '… average arithmetic development does not pursue a straight, fully predictable course of acquisition, but rather can be characterized by quite impressive individual differences'.

For example, Cockcroft (1982) reported that an average British class of 11-year-olds is likely to contain the equivalent of a 7-year range in arithmetical ability. Despite many changes in UK education since then, including the introduction of a standard National Curriculum and a National Numeracy Strategy, almost identical results were obtained by Brown et al. (2002). They found that the gap between the 5th and 95th percentiles on standardized mathematics tests by children in Year 6 (10- to 11-year-olds) corresponded to a gap of about 7 years in 'mathematic ages'. Individual differences in arithmetic among children of the same age are also very great in most other countries, though some studies suggest that they are less pronounced in Pacific Rim countries (Mullis, Martin, Goh, & Cotter, 2015).

Individual differences in arithmetic are easier to detect in *children*, who are attending school and taking mathematics tests, but it is clear that they persist throughout life. Some adults have severe difficulties with basic numeracy, and most such adults were already struggling with arithmetic by the age of 7 (Bynner & Parsons, 1997; Parsons & Bynner, 2005). By contrast, others have a fascination with numbers, are exceptionally skilled calculators and/or reason

exceptionally well about numbers. A few are 'lightning calculators' who can perform multi-digit calculations in their heads as fast as or faster than the average calculating machine.

Environment, heredity and arithmetical ability

There are numerous environmental influences on arithmetical development. Culture, education and even the language that we speak all play an important part in the way that arithmetic is acquired. These influences will be discussed in Chapter 9 of this book. More rarely, illness or injury can cause brain damage, which has a significant effect on arithmetic. Perinatal factors, such as severe prematurity, can have gross or subtle effects on brain development which influence arithmetic (Isaacs, Edmonds, Lucas, & Gadian, 2001; Simms et al., 2015; Tasaka & Shimada, 2000). The effects of brain damage and unusual patterns of brain development will be discussed in Chapter 10.

As regards genes, there are a few genetic disorders, such as Turner's syndrome and Williams syndrome, which are particularly associated with arithmetical difficulties. Such disorders are rare, but there is a tendency for arithmetical difficulties to run in families, even when not associated with a known genetic disorder. For example, Knopnik, Alarcon and DeFries (1997) found that monozygotic twins were significantly more likely than dizygotic twins to be concordant for mathematical disabilities. There was also a significant genetic correlation between mathematical disabilities and reading disabilities.

The possible genetic contribution to arithmetical disabilities will be discussed further in Chapter 10. There has been a relatively small, but increasing, amount of research into the possible genetic contribution to variation in arithmetical performance in the population as a whole.

Heritability is defined as the proportion of variance in a particular characteristic for a given population that can be explained by heredity. It is usually investigated by comparing levels of similarity in identical versus non-identical twins and/or in biological versus adopted relatives. There is considerable controversy both about the accuracy and appropriateness of the methods used to estimate heritability, and about the validity of even attempting to separate genetic from environmental contributions to individual differences, since there are undoubtedly strong interactions between genetic and environmental factors.

Bearing these limitations in mind, the relevant studies suggest that there is a significant genetic contribution to individual differences in mathematics. Vandenberg (1966) carried out a twin study and obtained a heritability estimate for mathematical ability of approximately 50%. More recent twin studies have also suggested significant heritability for mathematical ability (Davis, Haworth, & Plomin, 2009; Hart, Petrill, & Kamp Dush, 2010; Kovas et al., 2013; Oliver et al., 2004). Heritability estimates in the various studies range from 0.32 to 0.72, seemingly dependent on the tasks used and on the

country where the research is being done. On the whole, British studies tend to give higher heritability estimates than American studies. This may possibly be because Britain has a National Curriculum, and thus the environment is more uniform with regard to mathematical experiences, meaning that a lower proportion of variance is due to environment (Tosto, Haworth, & Kovas, 2015).

Further evidence for some genetic component to mathematical difficulties comes from a study of siblings of children with mathematical disabilities. Desoete, Praet, Titeca and Ceulemans (2013) compared 9 siblings of children with diagnosed mathematical learning difficulties with 63 age-matched children with no family members with known mathematical learning difficulties. They were all given three number line estimation tests involving sets of (1) dots, (2) numerals and (3) number words. The siblings of the children with mathematical difficulties performed significantly worse than the other children on all these tests, especially those involving numerals and number words. Although one cannot rule out some environmental influences affecting children in the same family, the study does suggest that those who are genetically related to those with mathematical learning difficulties are likely to have relative weaknesses in numerical abilities, even if they do not always amount to full-blown mathematical disabilities.

The major debates about genetic and environmental influences on mathematical ability have tended to focus not so much on population variance as on *gender differences*. Although gender differences are not in fact great in most aspects of mathematics, they have been an intense focus for research in the field of individual differences in mathematics. This topic must therefore be considered at some length.

Gender differences in arithmetic

"Girls [or boys] go to Mars
To get more stars.
Boys [or girls] go to Jupiter
To get more stupider."

(British playground rhyme, reversible according to the gender of the speaker.)

> I do wish I could do clock sums and those sums about taps running in and out. I wonder whoever invented such nonsense. I wish I could give the man who thought of them my honest opinion of his wits.
>
> How do you know it was a man?' inquired my brother … 'It is girls who think of ridiculous things.
>
> Not sums, especially pipe sums. What girl would let water in by two taps and out by one if she wanted to fill a bath? It takes a boy to think of that way of doing things.
>
> *(Maude Forsey: Mollie Hazeldene's Schooldays; Nelson, 1924, p. 16)*

Are males better at mathematics than females?

The folk wisdom is that males are better at mathematics than females. There is little evidence for this in the population as a whole. Groups selected as being to be *extremely* good at mathematics do tend to be predominantly male (Benbow, 1988). Certainly, significantly more males than females are to be found in advanced undergraduate mathematics courses, and the large majority of professional research mathematicians are men. However, Spelke (2005) notes a significant increase over time in the number of American female undergraduates studying mathematics. At the other end of the scale, most studies indicate that serious mathematical difficulties are *equally* common in males and females (Gross-Tsur, Manor, & Shalev, 1996; Lewis, Hitch, & Walker, 1994), and Reigosa-Crespo et al. (2012) found that the combination of severe mathematical difficulties with problems in basic number representation was actually commoner in boys (see section in this chapter on 'Dyscalculia: what is it and is it a separate condition?').

Gender differences in mathematics have decreased overall over the year, as shown by the successive TIMSS (Trends in International Mathematics and Science Study) international studies of pupils' mathematical performance (Mullis et al., 2015; Mullis, Martin, Foy, & Hooper, 2016a; Mullis, Martin, & Loveless, 2016b). When international surveys combine data from many countries, there is still some advantage in secondary school in favour of boys, but this is small and is not found in all countries, and a few countries such as Jordan and Malaysia show an advantage for girls.

Spatial ability and gender

Spatial ability has frequently been suggested (e.g. by Casey, Nuttall, & Benbow, 1995) to be an important factor in mathematical performance in general, and in gender differences in mathematics in particular.

There is no doubt that gender differences in spatial ability are frequently observed, though they tend to be quite small and are mainly found for tasks involving three-dimensional than two-dimensional space (Voyer, Voyer, & Bryden, 1995). There has been much debate as to whether such gender differences are due to biological factors (Geary, 1996) or to environmental factors such as boys being given more constructional toys and more freedom to explore their environment independently. Can this explain gender differences in mathematics?

Relationships between spatial and arithmetical abilities will be discussed later in this chapter. Though there is much evidence for spatial representation being involved in arithmetic, it is still controversial whether there are strong links between three-dimensional spatial abilities and non-geometrical mathematical abilities.

It is sometimes suggested that there may actually be gender differences with regard to the extent to which spatial ability predicts mathematical performance, but findings are contradictory with regard to the nature of such gender

differences. Some studies (Casey, Pezaris, & Nuttall, 1992; Connor & Serbin, 1985) have found that spatial ability predicts mathematical performance more in males than in females, while the opposite was found by Friedman (1995).

Thus, if arithmetic is indeed influenced by gender differences in spatial ability, the influence is likely to be quite limited and subtle.

Attention, mathematics and gender

As suggested by Dowker (1996), future consideration of gender differences in mathematics should address the issue of single-minded concentration. Personal communications from several male and female mathematicians suggest that one important factor in mathematical research, to a greater extent than for other academic subjects, is the ability to concentrate single-mindedly on a given problem.

There is some evidence that women may be more able than men to pay attention to several topics at once, and less willing or able to focus attention in the exclusive way demanded, for example, by advanced mathematics. There are obvious social reasons why this may be the case: women in their traditional role as caregiver to children may need to be able to divide their attention between their own tasks and several children's different needs and requests. There may also be biological factors involved. Males are more likely than females to be diagnosed as having 'attention deficit disorder'. Though this condition is probably overdiagnosed in some places, and there are uncertainties as to the criteria for diagnosis, this difference does suggest that males may be more likely than females to be found at either extreme of the ability to focus attention narrowly.

Mathematics self-concept, mathematics anxiety and gender

As discussed in Chapter 11, mathematics arouses anxiety and other negative emotions in many people: often linked to low estimates of their own ability and expectations of failure. There is also evidence that females tend to experience more anxiety than males with respect to mathematics, and to rate themselves lower in mathematics than do males. Studies tend to show that boys *like* mathematics more than girls do, and that girls' attitude to mathematics declines more over time than that of boys. However, the gender difference in attitude may be less now than some years ago.

Our own research has found no overall gender differences in primary school children's liking for mathematics (Dowker, Bennett, & Smith, 2012; Chapter 11 of this book). However, girls do tend to give themselves lower ratings in mathematics than boys do.

It is often suggested that females' greater anxiety and lower self-concept in mathematics may lead to poorer performance and less inclination to choose to study mathematics.

The question is which comes first. Do females find mathematics more difficult for other reasons, and therefore experience more anxiety about it; or is the

difficulty due to the anxiety? If the anxiety comes first, then why are females more anxious? They may be more liable to performance anxiety in general, but then why is their literacy development not equally affected by anxiety? One possible explanation is that they experience more negative reactions to their mathematical performance from adults and peers as a result of social stereotypes. If a boy performs badly at a mathematics test, this may be attributed to a temporary cause: he was not feeling well on the day, or he did not understand particular questions, or he had not revised sufficiently for the test. If a girl performs badly at the same test, she may be more quickly labelled as 'no good at maths'.

> Of course, boys usually can't understand algebra!
>
> *(Remark made by a female teacher to the parents of a 14-year-old boy, Britain, 2002)*

In Britain, some of the gender-related expectations about mathematical performance may have diminished or even reversed in recent years. As boys' general academic performance has declined in comparison with girls, teachers and policymakers have become increasingly concerned with male underachievement. Until the 1980s, books and papers on gender issues in education tended to be concerned mainly with the disadvantages experienced by girls. Since the 1990s, there have been far more writings on the educational difficulties and disadvantages experienced by boys. It would be of interest to investigate whether this has changed adults' perception of gender differences in mathematics. Anecdotal evidence – e.g. the remark that heads this paragraph – suggests that this is a possibility, though gender stereotypes about females being worse at mathematics are still prevalent (see Chapter 11 for more discussion of such stereotypes and their influences on mathematics anxiety).

People with difficulties in arithmetic

There will be many references in this book to children and adults with 'difficulties' in arithmetic, mathematics or numeracy. The term simply refers to children or adults who struggle or fail to cope with some of the aspects of arithmetic which are necessary or desirable for educational or practical purposes. Since the term 'learning difficulties' is used in some educational and clinical contexts to imply severe and unusual problems that may require special education, it should be emphasized that the use of the term 'difficulties' does not have such an implication here. It does not mean that these difficulties are pathological, that they are always classifiable as 'specific learning difficulties' in the technical sense or that they must be either innate or the result of brain damage. Some difficulties with arithmetic do indeed involve a pathological process, as in the case of adults who lose previously established arithmetical concepts or skills as the result of brain damage. However, most difficulties in arithmetic, like most difficulties in learned subjects, lie on a 'normal' continuum between extreme talent and

extreme weakness, and are due not to brain damage but to a mismatch between an individual's pattern of cognitive strengths and weaknesses and the way that (s)he is taught.

Dyscalculia: what is it, and is it a separate condition?

In recent years, there has been increased emphasis on the possibility that, just as some children have dyslexia, some may have dyscalculia: a specific difficulty in doing arithmetic (Butterworth, Varma, & Laurillard, 2011; Kuhn, 2015; Landerl, 2013).

There are a number of studies that investigate the nature and prevalence of specific learning difficulties in mathematics. For example, Lewis et al. (1994) studied 1,056 unselected 9- to 10-year-old English children (the entire age group within a particular, socially highly heterogeneous, local education authority, excluding only those assessed as having severe general learning difficulties). They were given Raven's Matrices IQ Test, Young's Group Mathematics Test and Young's Spelling and Reading Test. 1.3% of the sample had specific arithmetical difficulties, defined as an arithmetic scaled score of 85 or below, despite a Raven's IQ score of 90 or above. A further 2.3% had difficulties in both reading and arithmetic (scaled scores of 85 or below in both the reading and arithmetic tests) despite a Raven's IQ score of 90 or above. Thus, the prevalence of arithmetic difficulties in children of at least average cognitive ability was 3.6%. The children with arithmetical difficulties were equally divided as to gender, which contrasts with the general finding that boys are far more likely than girls to have language and literacy difficulties.

Gross-Tsur, Manor and Shalev (1996) assessed the incidence of dyscalculia in a cohort of 3,029 Israeli 11- to 12-year-olds. The 600 children who scored in the lowest 20% on a standardized citywide arithmetic test were selected for further testing. 555 were located and given an individualized arithmetic test battery previously constructed and standardized by the authors. This included reading, writing and comparing numbers; comparing quantities; simple calculations; and more complex (multi-digit) calculations. 188 children or 6.2% of the total were classified as having dyscalculia, using the criterion of a score equal or below the mean for children 2 years younger. 143 of these children were located and received parental consent for further testing. This included the WISC-R IQ test (Wechsler, D. (1974). Wechsler intelligence scale for children—revised. New York: Psychological Corporation), and reading and spelling tests standardized on 70 age-matched typically developing children. Three children were excluded from the 'dyscalculic' group because they obtained IQ tests below 80. Of the 140 dyscalculic children, 75 were girls and 65 were boys, once again indicating an approximately equal gender distribution. Their IQs ranged from 80 to 129, with a mean of 98.2. They were assessed for symptoms of other learning problems. The researchers diagnosed 17% as dyslexic and 26% as having symptoms of attention-deficit hyperactivity disorder. They came from significantly lower

socio-economic backgrounds than the children without dyscalculia. 42% had first-degree relatives with specific learning disabilities.

Bzufka, Hein and Neumarker (2000) studied 181 urban and 182 rural German third-grade pupils. They were given standardized school achievement tests of arithmetic and spelling. 12 children in each sample (about 6.6% of the whole population) performed above the 50th percentile in spelling, but below the 25th percentile in mathematics. When the urban and rural children were compared, they showed little difference in the incidence of specific spelling or mathematics difficulties, but the urban children (who were on the whole of lower socio-economic background) were far more likely than the rural children to have difficulty with both (48.6% versus 3.3%).

Dirks, Spyer, Van Lieshout and DeSonneville (2008) carried out a study in the Netherlands, where they found that about 10% of children met criterion for mathematics learning difficulties, if the criterion was that they should be below the 25th percentile on a mathematics test and above the 25th percentile on a test of language or reading. The prevalence went down to 5.6% when the 10th rather than the 25th percentile was taken as the criterion.

Other studies have given estimates ranging from 3.4% to 13.8% (Bahramand, 2008; Barbaresi, Katusic, Colligan, Weaver, & Jacobsen, 2005; Geary, Hoard, Nugent, & Bailey, 2012; Ramaa & Gowramma, 2002; Santos, 2017).

Thus, the incidence of mathematical learning difficulties differs widely between studies, depending on the methods and criteria used. For example, they have used different IQ tests; different mathematics tests, which may be emphasizing quite different components; and different cut-off points for establishing deficit in both IQ and mathematics. Moreover, it has been pointed out (Desoete, Roeyers, & DeClercq, 2004; Kaufmann et al., 2013; Mazzocco & Myers, 2003) that findings about the incidence, nature and outcomes of mathematical difficulties may vary considerably, depending on whether one uses the criteria of *discrepancy* between mathematical performance and IQ, *severity* of mathematical weaknesses or *persistence* of mathematical weaknesses. Given the marked differences in criteria between studies, it is perhaps inappropriate to focus too much on the exact numbers obtained. The main conclusion one can gain from most studies is that many children have difficulties with mathematics, and a significant number have relatively *specific* difficulties with mathematics.

Can some or all of the individuals with mathematical difficulties be described as having a specific condition that may be appropriately termed 'dyscalculia'? This question may be interpreted in several ways: Do a significant number of people have global deficits in arithmetic and not in other areas? Do some people have deficits in numerical abilities that are not part of a continuum, but put them in a sharply distinct group? If so, is there a single cause for such deficits?

It appears unusual for children to have difficulties in all areas of arithmetic, and at the same time no significant difficulties in other areas. As emphasized throughout this book, arithmetic is made up of many components. On the whole, children tend to have difficulties in only some of these components,

rather than across the board of arithmetic (Cowan et al., 2011; Dowker, 1998; Gifford & Rockliffe, 2008); or if they do have global arithmetical difficulties, these tend to be associated with other problems as well. For example, Gifford and Rockliffe (2008, p. 21) found that in one group,

> no pure cases [of dyscalculia] were found, although the children presented complex patterns of learning difficulties and compensatory strategies. The range of contributory factors suggests the need for new theoretical perspectives to consider learning difficulties and the need to study individual mathematics learning trajectories.

One cannot, however, rule out the possibility that a subset of people with mathematical difficulties do have a more specific deficit describable as 'dyscalculia'. For example, Reigosa-Crespo et al. (2012) studied 11,652 children in Havana, Cuba, from 2nd to 9th grades. 3.4% of this population had deficits in basic numerical capacities, such as recognizing and comparing small quantities, and almost all of these showed calculation difficulties. An additional 9.35% showed poor calculation, without the basic numerical deficits. Those with basic numerical difficulties differed in some important ways from the rest of the children with poor calculation. For example, the former problem was four times as common in boys as girls, while the latter showed no gender differences. Reigosa-Crespo et al. (2012) suggest that only those with the deficits in basic numerical skills should be described as dyscalculic. Clearly, there is much more research to be done on the definition of dyscalculia, and how it fits into the larger group of children with mathematical difficulties and the broad range of arithmetical abilities.

It is, in any case, questionable whether one can or should even attempt to establish a precise figure for the incidence of mathematical difficulties. If dyscalculia implies an impairment in all aspects of arithmetic, and *only* in arithmetic, then it would appear to be very rare (Dowker, 1998; Gifford & Rockcliffe, 2012): probably occurring in far fewer than 6% of the population. If it implies difficulties with certain aspects of arithmetic, which are sufficient to cause significant practical and educational problems for the individual, then they are probably considerably more frequent than the 6% figure would imply: perhaps occurring in 15%–20% of the population if not more. The exact figure is much less important than the fact that there are a large number of people who find arithmetic extremely difficult, fall increasingly behind in it during their school years, often come to fear it and as adults are restricted by the need to avoid jobs and other activities that involve arithmetic.

Can dyscalculia be defined?

Some argue that dyscalculia is not a single entity. Karagiannakis and & Coorman (2014) point out that there are several different potential causes for mathematical difficulties: e.g. difficulties in core number skills, visual-spatial difficulties, language

difficulties, reasoning difficulties and memory difficulties. They conclude that it is better to speak of mathematical learning difficulties, for which interventions may vary according to the causes, than of a single 'dyscalculia'. A possible counterargument is that perhaps there is a single entity that can be called dyscalculia, but that the term should be restricted to those problems that are caused by a specific deficit in core number skills. But even with this restriction, mathematical difficulties are quite diverse, given that arithmetical thinking involves a very wide variety of components (see Chapter 2). Thus, there are many forms and causes of arithmetical difficulty, which may assume different degrees of importance in different tasks and situations.

Numeracy difficulties in adults

There are considerable individual differences in arithmetical performance in adults. Some adults have a considerable facility with number and enjoy it, and others find arithmetic very difficult. Because of the practical problems that may be caused by numeracy difficulties, the lower end of the adult performance range in arithmetic has received rather more attention than the higher end.

Bynner and Parsons (1997) and Parsons and Bynner (2005) gave some Basic Skills Agency literacy and numeracy tests to a sample of 37-year-olds from the National Child Development Study cohort (which had included all individuals born in Britain in a single week in 1958). The numeracy tests included such tasks as working out change, calculating area, using charts and bus and train timetables, and working out percentages in practical contexts. According to the standards laid down by the Basic Skills Agency, nearly one-quarter of the cohort had 'very low' numeracy skills that would make everyday tasks difficult to complete successfully. This proportion was about four times as great as that classed as having very low literacy skills. Women were more likely than men to be in this category: 27% of women, as compared with 19% of men, were classed as having very low numeracy skills.

Poor numeracy did indeed seem to have an impact on these people's lives. People who had *both* poor numeracy *and* poor literacy skills had the greatest problems in employment. However, even if literacy was adequate, people with poor numeracy skills had their problems. Among men, 17% of those with only numeracy difficulties were not in full-time employment, as compared with 10% of those with only literacy difficulties, 8% of those with problems in neither area and 31% of those with problems in both areas. Among women, 74% of those with only numeracy difficulties were not in full-time employment, as compared with 58% of those with literacy problems only, 56% of those with problems in neither area and 73% of those with problems in both areas. Among those who were in full-time or part-time employment, the *type* of job was associated with numeracy skills. Among men, 74% of those with only numeracy difficulties were working in a manual occupation, as compared with 69% with only literacy difficulties, 39% of those with problems in neither area and 75% of those with problems in

both areas. Among women, 41% of those with only numeracy difficulties were working in a manual occupation, as compared with 18% with only literacy difficulties, 25% of those with problems in neither area and 55% of those with problems in both areas. Although the direction of causation may not be completely one way – certain forms of occupational training and experience may serve to increase numeracy skills – it does appear that poor numeracy skills do affect people's ability to gain full-time employment, and often restrict employment options to manual (usually lower-paid) jobs.

The effect of numeracy skills on income was indeed quite significant. Even if only people who left school at 16 are considered (to control for the effects of extended education on earnings), 40% of men and 58% of women with only numeracy difficulties were classed as receiving low wages, as compared with 35% of men and 30% of women with only literacy difficulties, 43% of men and 55% of women with problems in both areas, and 20% of men and 22% of women with problems in neither area. Moreover, it seems that poor numeracy skills have, if anything, a *greater* impact than poor literacy skills on employment prospects; however, this may be in part because literacy skills did not need to be as severely deficient as numeracy skills to enable people to be rated as having a difficulty in the area.

Does early numeracy performance predict later numeracy performance? There is certainly evidence that adults with numeracy difficulties tend to have already demonstrated numeracy difficulties in childhood. The majority of the adults with numeracy difficulties in Bynner and Parsons' (1997) study had already been rated by their teachers as poor at arithmetic at the ages of 7 and 11. However, the relationship was not perfect. Fifteen per cent of the adults with numeracy difficulties had been rated as outstanding at arithmetic by their teachers at the age of 7. It seems that while numeracy difficulties are often consistent throughout life, some people start out with no numeracy problems, but then 'something goes wrong' during their development and education.

Severe, and slightly less severe, specific difficulties with arithmetic

There are some individuals who have very severe difficulties with most aspects of arithmetic, while having no apparent difficulties in other areas. Butterworth (1999) described "Charles", a university graduate, who had no problems with literacy or general reasoning, but who could only solve even single-digit sums by counting slowly on his fingers. He could not subtract or divide at all, or carry out any sort of multi-digit arithmetic. He was extremely slow even at comparing numbers: for example, saying which was bigger, 9 or 3. He could only give the answer to such comparison problems after counting on his fingers from the smaller number to the larger number. He took about 10 times as long as most people even to state whether two numerals were the same or different. He seemed to lack even the most basic numerical abilities that usually seem to be present in babies: he could not even recognize two dots as two without counting them.

Chapter 10 will discuss some of the possible brain-based reasons for the difficulties of people like Charles. For now, such cases will be treated as evidence that it is possible to have very severe arithmetical difficulties without having difficulties in other aspects of language, memory or reading.

Such extreme cases are rare, even among those with diagnosed dyscalculia. Nicola is perhaps more representative. She was a 20-year-old woman who had been diagnosed as having dyscalculia. She had always had severe difficulties with school arithmetic. Her Wechsler Adult Intelligence Scales (WAIS) Arithmetic subtest score was 2. On the British Abilities Scales Basic Number Skills subtest, designed for children, she obtained an age equivalent score of 8 years 9 months. This test places considerable emphasis on the ability to read two- and three-digit numbers, which Nicola could do. It is possible that a test that places less emphasis on number reading would have resulted in an even lower age equivalent. She dealt with most arithmetical problems by counting on her fingers: counting on from the larger number for addition and counting down from the larger number for subtraction. What were her more basic numerical difficulties like? Her ability to estimate numerosities was indeed relatively weak, but she *could* recognize quantities up to 3 without difficulty. There was a sharp division for her between 3 and more than 3: for larger quantities, she counted on from 3, so that six dots are counted: "3, 4, 5, 6".

Thus, Charles and Nicola seem to represent quite different forms of dyscalculia. Charles lacks even some of the numerical abilities that appear to be possessed by most babies. Nicola has these numerical abilities, but has difficulty with most forms of arithmetic, and relies excessively on somewhat cumbersome counting strategies.

It is important that people like Charles, and even like Nicola, be identified as having difficulties early on, so as to reduce the risks of intellectual confusion and emotional frustration and possibly humiliation if they are expected to cope with the typical school arithmetic curriculum without special help.

Arithmetical difficulties: a difference in kind or a difference in degree?

One question which may be asked about people who have difficulties in any area of functioning is whether they are fundamentally different from individuals without such difficulties or whether they represent the lower end of a continuum. Do people with arithmetical difficulties lack some function which everyone else has, or are they doing the same things as others, but less well or less efficiently? Or are most people with arithmetical difficulties on the lower end of a continuum, while a subgroup – perhaps the ones who can be defined as 'dyscalculic' – have fundamental differences from the rest of the population?

It is difficult to make generalizations about *all* people with arithmetical difficulties, because, as we have seen, such difficulties take very heterogeneous forms. However, the evidence is that the majority of such difficulties can be seen as representing the lower end of a continuum. Studies of unselected groups

of individuals have indicated that individual variations are considerable even for adults in even such apparently basic numerical skills as counting accuracy and speed (Deloche et al., 1994) and single-digit addition and multiplication (Lefevre, Smith-Chant, Hiscock, Daley, & Morris, 2003; see also Chapter 4 of this book). One study by Szanto (1998) found that adults with arithmetical difficulties performed in a similar fashion on both computational skills and arithmetical reasoning to normally achieving adolescents and children who were at a similar overall arithmetical level.

This does not mean that people with arithmetical difficulties do not have a genuine problem. Many characteristics vary continuously in the population, and yet may pose serious problems at the extremes. Body weight varies continuously, but being very overweight or underweight may both indicate and cause serious health problems. IQ varies continuously, but those at the low end of this continuum are likely to experience major difficulties, even when no specific pathological cause can be identified. However, it does make it likely that most people with arithmetical difficulties are not sharply distinct from other individuals, and that most of the difficulties that they show can also be seen to varying degrees in the general population.

Developmental dyscalculia needs to be evaluated in the context of the significant individual differences that are found in the general population, and also the findings, reviewed throughout this book, that show arithmetical ability to be made up of numerous components which are all subject to individual differences that continue from childhood to adulthood. Such individual differences call into question theories of a single core deficit (Dowker, 2015; Kaufmann et al., 2013), unless we assume that only a small group of people, with mathematical difficulties associated with one specific deficit, should be termed dyscalculic, and that the majority of people with mathematical difficulties do not come under this umbrella (Reigosa-Crespo et al, 2012).

Exceptionally high mathematical ability

> To be a scholar of mathematics, you must be born with talent, insight, concentration, taste, drive and the ability to visualize and guess.
>
> *(Paul Halmos: I Want to be a Mathematician; Springer, 1985)*

Much work on individual differences in arithmetic has focused either on the 'normal' range of ability or on the nature of and reasons for arithmetical difficulties. However, there are of course some people who have exceptional talent in arithmetic. These may be approximately divided into two groups: those who are unusually fast and accurate calculators, and those who have exceptional ability to reason about numbers and arithmetic. It is possible for one individual to combine both types of ability to an exceptional degree, e.g. the mathematician and talented calculator, Professor Aitken, described by Hunter (1962). Such a combination seems, however, to be rare.

Extremely superior calculation skills

Arithmetical skill is usually associated with high levels of ability in other directions, but it does not depend on such levels of ability.

Extremely superior calculation abilities sometimes, though rarely, occur in people with very low IQs (Cowan, O'Connor & Samella, 2003;Hermelin and O'Connor, 1991; O'Connor and Hermelin, 1984; O'Connor, Cowan, & Samella, 2000; Smith, 1988). Some such individuals are globally outstanding at most or all aspects of calculation. More often, they are remarkably rapid and accurate at some particular aspect of calculation: surprisingly often, the ability to work out on which day of the week a particular date will fall. Usually, such special calculation skills are associated with a generally relatively good level of arithmetical performance. Most, but not all, of these 'savant' calculators are autistic.

Some people who are not intellectually impaired or autistic are capable of outstandingly rapid and accurate calculations. There have, for example, been a number of professional and semi-professional performers who earned fame and sometimes money by giving public displays of their arithmetical abilities (Flansburg, 1993; Smith, 1988; Weinland, 1948). Such talents have also been recorded in some people working in solitary occupations, such as the shepherds Jacques Inaudi and Henri Mondieux (Smith, 1988).

It is hard to say how frequently such talents occur in the general population. Since the advent of calculators and computers, which will perform rapid calculations, there are few practical advantages in being a calculating prodigy. Even people who are capable of such calculations may thus prefer to devote their major efforts to other areas, if not restricted in doing so by their own limitations (as in autism) or their circumstances (as in the isolated shepherds). Cowan, Stainthorp, Kapnogiann and Anastasiou (2004) studied two non-autistic primary school boys who demonstrated savant-type abilities in calendrical calculation. One was of average IQ, while the other had an IQ of over 140. Both were very good at other aspects of arithmetic. They had a wide variety of other interests. By the age of 10 or 11, they seemed to have lost interest in calendrical calculation, though they were still able to demonstrate it if required.

Such talents as calendrical calculation may even be seen as suspect by some people. George, a humanities graduate, informed me that at the age of 5 he had been very interested in and good at calendrical calculation, though the extent of his ability in this direction was never recorded. His parents, aware that this is sometimes a 'savant' talent in people with low IQs, had actively discouraged him from pursuing this interest, and he felt that this had not only led to the loss of his calendrical calculation ability, but reduced his interest in mathematics as a whole.

The relative lack of value placed on rapid calculation in contemporary Western society almost certainly means that some calculating prodigies go unnoticed and are never recorded. Does it also prevent people from becoming calculating prodigies in the first place? Would many more people be rapid calculators if they were encouraged to practice this activity? It does seem that most or all calculating

prodigies, whether 'savants', shepherds, public performers or the rare professional mathematicians in this category, do devote a lot of time to practicing their skill.

There is evidence that some people who might not otherwise have been outstanding calculators can develop an impressive skill in this direction with sufficient training and practice. Such skills do seem to be commoner in cultures where they have a practical or social value. Binet (1894) reported the lightning calculation abilities of some French cashiers before the days of calculators: the fastest cashier whom he studied could multiply two three-digit numbers in 4 seconds. In contemporary Japanese culture, though calculators have removed the practical necessity for rapid calculation, the ability to use an abacus for rapid calculation is still highly valued. Some teenagers join abacus classes and clubs and devote themselves seriously to this pursuit (Hatano, Miyake, & Binks, 1977). Many of these people become very rapid calculators, able to create a mental image of an abacus which they can use almost as efficiently as a real abacus (see Chapter 9 for more details).

Staszewski (1988) taught a group of American students some fast calculation strategies and gave them about 300 hours of training over a period of 2 or 3 years. After this extensive training, they were able to calculate such problems as 59,451 × 86 in only 30 seconds.

Such findings have led some people such as Howe (1991) to conclude that talent in arithmetic and other domains is just a matter of practice. According to this view, anyone could be a calculating prodigy if they had the time and the will to devote large amounts of time to practicing this skill. Savant calculation abilities, according to such theories, are a product of the practice induced by having very little else to do with one's time, combined with the obsessional characteristics of many autistic people.

Yet, though practice is important, it does not seem to be the whole story. It does not seem that *anyone* could be a lightning calculator. What if the people who have real difficulty in remembering and applying calculation facts and procedures? People who receive extensive training and practice in calculation are usually not selected at random. Not everyone in Binet's France chose to become a cashier, and presumably not everyone who did apply for a cashier's post was successful in obtaining the job.

In Japan, only about one-sixth of secondary school pupils choose to devote themselves seriously to learning the abacus. Staszewski's students were not *compelled* to devote hundreds of hours to calculations, although they did receive some payment for it: a person with poor initial calculation ability might not have chosen to engage in such activities or might have given up at an early stage.

High levels of arithmetical reasoning ability

Expertise at calculation procedures is not the same thing as the ability to understand and reason about numbers. The two may go together, but often do not. There have been several studies of the characteristics of children and adults

with high levels of mathematical reasoning ability. Most such studies have not distinguished between high levels of reasoning about numbers and high levels of reasoning about other aspects of mathematics.

Arithmetical reasoning ability often shows itself quite early in life. Not every child with a strong interest or talent in arithmetic will continue equally interested, or equally highly achieving, later on. For one thing, as children grow older, they have more other intellectual activities to choose from. Young children in many cultures are exposed to numbers, but it is only later that they are typically exposed to history or geography or the sciences or music. Thus, children with high IQs and/or intellectual interests are often interested in arithmetic at an early stage and transfer their interests to other spheres later on. For another thing, many children become less interested in intellectual activities in general as adolescence approaches: the peer group; the opposite sex; their appearance; and the pursuit of physical, social and economic independence all take priority. Also, secondary school mathematics may not have as broad appeal as primary school mathematics: a young child who is fascinated with numbers is not necessarily destined to be equally fascinated with algebra, calculus or proofs. While the reverse can also be the case – a child who dislikes numbers and arithmetic may later find far more interest in more abstract mathematics – it is probably less common. Children of all levels of mathematical ability tend to show a reduction in liking for mathematics as they get older (see Chapter 11).

There have been several studies of people with high levels of mathematical reasoning, though arithmetical reasoning is typically not treated separately from reasoning about other aspects of mathematics. For the most part, studies of 'mathematically able' individuals, whether children, adolescents, or adults, have grouped all such individuals together. Those with high numerical and arithmetical ability have usually not been studied separately from those with talents in geometry and other domains of mathematics.

The best-known study in the area is probably that carried out by Krutetskii (1996) in what was then the Soviet Union. The most important characteristics of the mathematically gifted children were as follows:

They *liked* mathematics and were enthusiastic about it.

They could reason quickly: sometimes to a degree which interfered with their ability to 'show their working' in written form or describe it clearly to others.

They were good at *generalizing*: they could not only find appropriate rules and strategies for solving specific problems, but could often work out general rules that could be applied to a broad category of related problems.

They could recognize the *mathematical structure* of problems.

They were able to deal with *abstract concepts*.

They were inclined to remember not the *details* of a problem, but its *general principles*. If they were shown another problem based on the same general principles as a previous one (e.g. successive problems that involved the inverse

relationship between multiplication and division), they often thought that they had seen the same problem before.

They were *flexible* in their thinking and were able to devise a variety of strategies and to move easily from one type of strategy to another.

Sheffield (1994, 1999) studied mathematically able children in the United States of America and devised a somewhat similar list of characteristics. These included *interest* and *curiosity* about quantitative information; the ability to perceive and generalize *patterns and relationships*; *flexibility* of thought; the ability to reason *analytically, inductively and deductively*; the ability to *vary* and switch between strategies; the ability to *transfer* learning to new situations; a tendency to *formulate* new mathematical questions; and *persistence* with difficult problems. These characteristics are very similar to those discussed by Krutetskii (1976). In addition, Sheffield (1994, 1999) stated that mathematically gifted children were good at organizing and working with *data*, and at distinguishing relevant from irrelevant data.

These characteristics were not specific to arithmetic, and not all of the mathematically able pupils in the studies by Krutetskii and Sheffield were necessarily very accurate calculators; indeed, Sheffield pointed out that some were not. Hope and Sherrill (1987) looked more specifically at children who were skilled at mental calculation. Such children were once again characterized by interest in the subject, flexibility of approach; and ability to perceive and make use of mathematical principles and relationships (in this case, arithmetical principles and relationships between numbers and operations). Memory for number facts was of secondary importance.

The importance of flexibility and variability of strategy use: central to mathematical talent and central to mathematical development

> There are nine and sixty ways of constructing tribal lays
> And every single one of them is right.
>
> *(Rudyard Kipling: In the Neolithic Age)*

A central theme of this book is that people use a very wide variety of strategies in arithmetic. In this book, this issue will come up repeatedly. For example, it will be discussed with regard to the use of 'derived fact strategies' based on arithmetical principles such as commutativity and associativity (Chapter 6); estimation strategy use (Chapter 7); and to the ability to represent the same problem in multiple formats – concrete, numerical and verbal – and to translate between them (Chapter 5). It is indeed an issue that permeates all aspects of arithmetic. There appears to be no form of arithmetic – from counting to complex arithmetical reasoning – for which people fail to use a remarkable variety of strategies.

This is an issue that can be obscured for at least two reasons. One reason is that oversimplified interpretations of stage theories of development can lead to an assumption that children at a given stage use only one strategy and then move on to using another single strategy at the next stage. In fact, most stage theorists including Piaget do not actually state this. However, the emphasis on changes with strategy according to the stage of development can sometimes obscure the fact that multiple strategies are often used by the same child within the same 'stage'.

The other reason is that educators and psychologists often note the obvious fact that some strategies are more accurate and efficient than others, and leap from that fact to the conclusion that there must be one strategy that is 'best' for any given problem, and which children should learn to use. This view may have received some impetus from attempts to create 'expert systems' in artificial intelligence. Often such research on 'expert systems' has been based on, and has reinforced, a view that expertise results in effective decision-making as to which is the best strategy to use for a given problem, and efficient use of that strategy.

In fact, even young children use many different strategies for similar arithmetic problems at the same time. As early as 1928, Alpert studied preschool children's responses to a number of problem-solving tasks and found that the same child would attack one problem in one way and another problem in a different way, and no one problem was solved in the same way by all children. More recently, studies by researchers such as Baroody (1988, 1989) and Siegler and his colleagues (Kerkman & Siegler, 1997; Siegler & Jenkins, 1989; Siegler & Robinson, 1982) have shown that 5-year-old children in the early stages of arithmetic use a multiplicity of strategies for simple addition problems and may solve exactly the same problem in different ways at slightly different times. Development consists not of the replacement of a single immature strategy by a single more mature strategy but of the discovery of increasingly more mature strategies, which coexist for a long time with immature strategies, before gradually supplanting them. Similar results are found for older children with very low IQs (Baroody, 1988; Fletcher, Huffman, Bray, & Grupe, 1998).

This multiplicity of strategy use is not solely due to the fact that young children have not yet worked out how to use a 'best' strategy. Young children and other novices may be variable in their strategy use, secondary pupils often are as well (Liu et al, 2018; Van Hoof et al., 2013), and mathematically able adults seem to be even more so – even when dealing with quite simple arithmetic. Some years ago, I gave mathematicians a dull-seeming task that involved estimating the answers to 20 multi-digit multiplication and division problems (Dowker, 1992). The details of the study are described in Chapter 7, but the findings and conclusions are so central to the global theme of this book that they must be mentioned here. The mathematicians were very accurate (though not error-free) estimators, but what was remarkable was the number and variety of the estimation strategies that the different mathematicians used for the same problem. This was not just due to individual differences in strategy preference (between-individual variability) but reflected the fact that the mathematicians were variable and versatile

in their strategy use, as shown most strikingly by the fact that when 18 of the mathematicians were retested after a few months, they used different strategies on the two occasions for most of the same problems.

A few years later, the study was extended to other groups of adults: accountants, psychology students and English students (Dowker, Flood, Griffiths, Harriss, & Hook, 1996; also see Chapter 7 of this book). The most striking result of the study was that all the groups in the study showed both between-individual and within-individual variabilities to a very marked degree. All groups used a remarkably large number and variety of strategies for every problem. Within-individual variability was also considerable, as shown both by the number of broad types of strategy used in each protocol and by the fact that those who were retested showed considerable variations in specific strategy use on the two test occasions.

Although great variability in this task was a characteristic of all groups, there were some highly significant individual and group differences. These were mostly in the direction of greater variability being associated with greater arithmetical knowledge and experience. The mathematicians showed more variable strategy use than the accountants or psychology students, who in turn showed more variable strategy use than the English students. Within each group, those who used more strategies tended also to be the more accurate estimators.

Such results indicate that the view of an 'expert' as one who is able to work out quickly how to use the 'one best strategy' is inaccurate. Either expertise is consistently associated with variable strategy use, or there may be a U-shaped relationship between expertise and strategy variability as suggested by Dowker et al. (1996). Novices may use many strategies, often inappropriate or inefficient, because they have not yet fixed on any particular strategy or a set of strategies. Experts may use many strategies, mostly appropriate, partly because they have access to more strategies, but mainly because they have a sufficiently good 'cognitive map' of the territory that they do not fear becoming irretrievably lost if they stray from a known path. People at intermediate levels of expertise are more likely to confine themselves to a small set of strategies which they have learned, and with which they feel safe. There may also be some link to the negative effects of *mathematics anxiety* on working memory and other aspects of executive functions (see Chapter 11). Strategy selection and flexibility are strongly linked to executive functions (see Chapter 5), and perhaps for that reason, Si, Xu and Liu (2011) found that people with higher mathematics anxiety perform worse at some computational estimation tasks. Mathematicians and other experts are unlikely to suffer from significant mathematics anxiety, both because their expertise is likely to increase confidence and because people with highly negative attitudes to mathematics are unlikely in the first place to become professional mathematicians or follow other careers that require the intensive study of mathematics.

The findings concerning the remarkable variability of strategy use by mathematicians in apparently simple arithmetic problems also received support from the subsequent findings by Lefevre and her colleagues (Lefevre et al, 2003; Lefevre,

Sadesky, & Bisanz, 1996a; Lefevre et al., 1996b) that educated adults use a wide variety of strategies in solving single-digit addition and subtraction problems. Smith (1995) found great variability in adults' strategy use in problems involving fractions and decimals. Obersteiner, Van Dooren, Van Hoof and Verschaffel (2013) found that mathematicians use a wide variety of strategies for comparing fractions.

Such findings concerning strategy versatility have important educational implications. If strategy variability even for simple problems is the norm, and becomes startlingly high in 'expert' groups, then it becomes obvious that there is *not* only one way or even one optimum way of solving a problem. This supports the view of Ginsburg (1977) that "it is nonsense to suppose that there is one dominant mode of learning academic material" and that it is desirable "to explore different kinds of learning for different kinds of children, or for the same child under different circumstances". There is now abundant evidence that elementary school children will, if permitted, devise and use a wide variety of appropriate strategies for addition, subtraction, multiplication and division (Ambrose, Baek, & Carpenter, 2003; Beishuizen, Van Putten, & Van Mulken, 1997; Fuson & Smith, 1997; Thompson, 1994, 1997a, 1997b).

Both as cause and effect of such findings, there has been an increased emphasis on multiple strategy use in school mathematics education (Star & Rittle-Johnson, 2008; Star et al., 2015). There is some evidence that teachers' encouragement of pupils' using multiple strategies does indeed lead to greater flexibility (Blöte, Van der Burg, & Klein, 2001; Star & Rittle-Johnson, 2008; Star & Seifert, 2006; Star et al., 2015). More work is now needed on the best ways of achieving this outcome, how long-term the effects are and whether they are similar at all ages.

Indeed, some attempts to model human mathematical cognition through artificial intelligence programs have emphasized flexibility and inventiveness in strategy use rather than honing in on one optimal strategy. Defays' (1995) 'Numbo' program is a notable example.

Do numerically gifted people make mistakes in arithmetic?

Professional academic mathematicians are one group who have been selected and self-selected for talent in at least some aspects of mathematics, though not necessarily arithmetic.

There are two common stereotypes about the arithmetical calculation ability of professional mathematicians and others with considerable capacity for abstract mathematical reasoning. One is that such people are 'human calculators' who can solve arithmetical problems with remarkable speed and accuracy. The other is that they are particularly bad at arithmetic and 'cannot even add their laundry lists'. Both stereotypes are inaccurate, except with regard to rare cases. Most mathematicians are more accurate in arithmetic than people with limited mathematical ability, but are far from immune to errors. For example, in the estimation study alluded to above and in Chapter 7, only 11 of the 62 protocols

by the mathematicians resulted in perfect scores. This is not just an artefact of the scoring technique, and the set criterion for an accurate estimate, because when mathematicians' strategies were inaccurate they tended to be *completely* inaccurate. Mistakes were generally due to calculation errors, most frequently involving a failure to adjust the decimal point.

Every one of the five mathematicians studied by Lewis (1981) made at least one error in solving 22 algebra problems, and most of these were "careless" errors in carrying out routine operations. Lewis suggests that "no one bothers to become a flawless equation solver". Mathematicians, who have written about the nature of their own mathematical thinking, do indeed emphasize that the avoidance of such errors is not their primary concern. Mary Cartwright (1955), for example, stated that

> … there is no doubt that even the greatest mathematicians make plenty (of mistakes). In fact it was stated of one eminent professor that he says a, he means b, he writes c, but it *should* be d. If the proof is easily rectified and the result holds, we do not worry.

Generally, even when the mathematicians were making use of learned facts about numbers, such as squares and square roots, their knowledge and their use of strategies based on this knowledge seemed to involve an enjoyment of thinking about and playing with numbers, rather than rote memorization.

Such interest in playing with numbers also characterizes many other people with high level of arithmetical reasoning ability. A similar attitude was described by some of Hope and Sherrill's (1987) talented mental calculators, and perhaps best expressed by the outstandingly able 13-year-old mental calculator studied by Hope (1987), who frequently used strategies based on factoring and on a remarkable knowledge of squares. She "remarked, 'I'm not very good at memorizing.' (Hope) commented that some people might regard her knowledge of squares as merely 'memorizing'. She exclaimed, 'But that's not memorizing. That's knowing and thinking!'"

What else might distinguish mathematically talented people from others?

So far, we have looked mainly at mathematically talented people mainly from the point of view of aspects of their arithmetical reasoning, flexibility in arithmetical problem-solving and interest in playing with numbers. Are there other characteristics that may distinguish mathematically talented people from others?

There have been several studies of the characteristics of mathematically gifted adolescents and young adults (Benbow, 1988, 1992; Lubinski & Humphreys, 1991; O'Boyle, 2000; O'Boyle, Gill, Benbow, & Alexander, 1994). Once again, most such studies have not separated those who are specifically gifted at dealing with or reasoning about numbers from those who are gifted at other aspects of

mathematics such as geometry. Moreover, the mathematically gifted pupils have typically been selected according to test scores or academic performance, which may have excluded some pupils, especially those who were underachieving at school (Sowell, Bergwall, Zeigler, & Cartwright, 1990). Their main findings are that mathematically gifted individuals tend to be male, to be short-sighted, to show a high rate of allergies and autoimmune disorders, and to show unusual patterns of brain organization with a high rate of left-handedness.

Benbow (1988, 1992) and her colleagues have suggested that high levels of testosterone may lead both to unusual patterns of brain organization conducive to mathematical talent and to a proneness to immune disorders. This theory, however, is very much in the realm of speculation. And as stated in the section on gender differences, the reasons for a higher proportion of males among mathematically gifted individuals are not at all clear.

Some studies have focused more on the domain-general cognitive characteristics of mathematically gifted children and adolescents. Logical reasoning abilities are important to advanced mathematics, which emphasizes abstract reasoning and logical proofs. Such reasoning abilities appear to be associated with mathematical giftedness quite early on. For instance, Morsanyi, Devine, Nobes and Szucs (2013) found that mathematically gifted children showed outstanding skills in logical reasoning about belief-laden problems.

In particular, spatial ability is frequently found to be higher in these individuals than in other pupils (Hermelin & O'Connor, 1986; Lubinski & Benbow, 2006; Myers, Carey, & Szucs, 2017; Van Garderen, 2016). Leikin, Paz-Baruch and Leikin (2013) found that verbal working memory was strongly related to generally high IQ as measured by Raven's Matrices Test, but that spatial working memory was more strongly related to high *mathematical* ability. Interestingly, one of the few studies carried out with young children produced somewhat conflicting results: Swanson (2006) found that 6- to 8-year-olds with high mathematical ability were characterized by good performance on tests of working memory, naming speed and the ability to inhibit irrelevant responses, but did not differ from other children in visual-spatial ability.

Myers et al. (2017) have pointed out the need for caution in interpreting results because most studies had small sample sizes, resulting in limited statistical power, and because there were many differences between studies as regards both the selection criteria for the mathematically gifted groups and the nature of the control groups: for example, whether they were gifted in other ways than in mathematics or selected for being average mathematical performers in the same schools.

As the mathematically able young people reach adulthood and select advanced mathematics as a topic of interest, domain-general characteristics, as well as some more mathematics-specific abilities, continue to be important. Wei, Yuan, Chen and Zhou (2012) found that in a sample of Chinese undergraduates, advanced mathematical performance (algebra, statistics and mathematical logic) was independently related to both spatial ability and language comprehension, and to

some higher-level numerical processing abilities, but not to basic numerical processing or computation.

Several studies have looked at cognitive characteristics of professional academic mathematicians. We have seen above that they are flexible and variable in their mathematical strategy use, and that they have strong positive emotional reactions to mathematics. They also seem to show certain differences from others in their representations of numbers. Cipora et al. (2016) compared doctoral students in mathematics with a control group in engineering and other subjects involving numeracy and another control group in humanities subjects. The controls showed the usual Spatial-Numerical Association of Response Codes (SNARC) effect of responding faster to smaller numbers with the left hand and larger numbers with the right hand, but the mathematicians did not. This may indicate that their representations of numbers are more abstract than those of most people.

Some studies have suggested that mathematicians have particularly high spatial skills. Sella, Sader, Lolliot and Cohen Kadosh (2016) found that mathematics graduate students and early career researchers scored higher in non-verbal, but not in verbal, IQ than controls in the humanities. They were also more accurate in mapping positive numbers onto an empty number line. However, the latter difference ceased to be significant after controlling for block design, which may indicate that apparent differences between mathematicians and non-mathematicians on basic numerical skills may be secondary to differences in visual-spatial processing skills. This is a very interesting finding, but caution may be needed in interpreting the results, as the mathematicians had similarly high verbal and non-verbal IQ scores, while the non-mathematicians showed a large discrepancy in favour of verbal score. Thus, it may be a matter of the humanities researchers showing relative weakness in spatial skills (perhaps leading to their choice of academic subjects that would exploit their high verbal skills while not requiring too much spatial ability), rather than the mathematicians being characterized by selectively high spatial ability.

The need for further research

More work needs to be done on arithmetical talent. If we are to study the entire spectrum of arithmetical ability, it is important to study the high end as well as the low end. Moreover, arithmetical talent is an interesting subject in its own right. However, most studies of mathematically gifted children and adolescents have not separated high numerical ability from other aspects of mathematics, let alone looked at different components of numerical ability. When studying mathematical giftedness in adults, the focus has tended to be either on the exceptional 'super-calculators', especially those who are 'savants', or on professional mathematicians. As we have seen, though most of the latter are arithmetically competent, their main interests tend to be in abstract logical reasoning and only a minority have a strong interest in numbers and their manipulation. There needs

to be more research on people with high ability and interest in arithmetic, who are nevertheless not 'lightning calculators', but may be more interested in arithmetical principles, and, like Hope and Sherrill's (1987) participant, in 'knowing and thinking'. With children, there need to be more studies such as that of Hope and Sherrill. As regards adults, there should be more studies of the people, who, for example, design Sudoku problems, or who, some years ago, competed on 'Countdown'. Although high arithmetical talent is not as linked to specific jobs as in the days of Binet's calculators, it still exists and is worthy of study. And at the higher as well as the lower levels of arithmetical ability, it is important to bear in mind that arithmetic is not a single ability but is made up of many components. This will be the theme of the next chapter and of the rest of the book.